8.9 Wheel Alignment

- It refers to the angular relationship between the front wheels and parts attached to it with that of car frame. Wheel alignments ensure that all four wheels are consistent with each other and are optimized for maximum contact with the surface of the road.

8.9.1 Aim of wheel alignment

1. To turn the steered wheel easily and to ensure vehicle directional stability.
2. To reduce the tyre wear to a minimum.

8.9.2 Angles in wheel alignment

1. Caster Angle
2. Camber Angle
3. King pin inclination
4. Toe-in and Toe-out

8.9.2.1 Caster Angle

The angle between the king pin centre line (or steering axis) and the vertical axis, when viewed from the side of the vehicle, is called the Caster angle. Caster angle is about 2° to 3°. As shown in figure 8.9.2.1 (a), (b), (c).

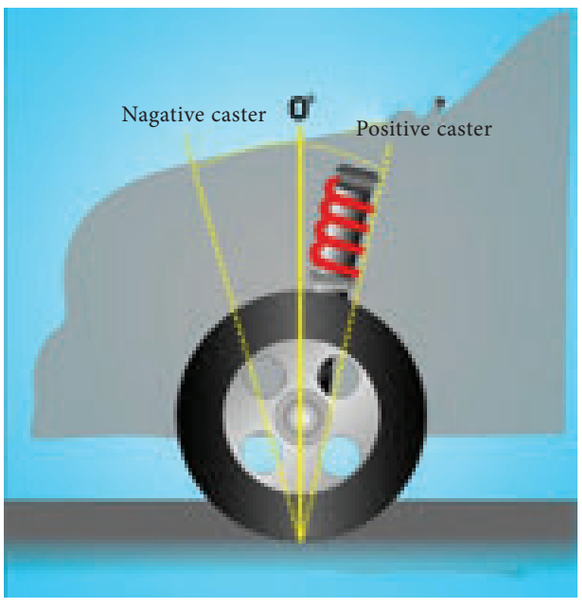


Figure 8.9.2.1 (a) Caster angle

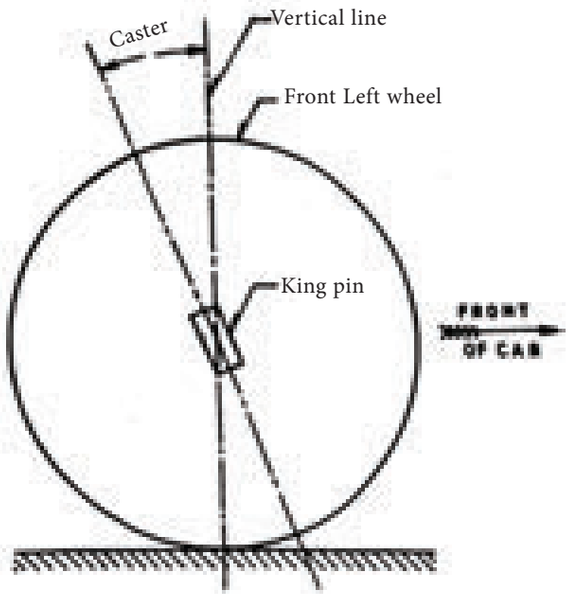


Figure 8.9.2.1 (b) Caster angle

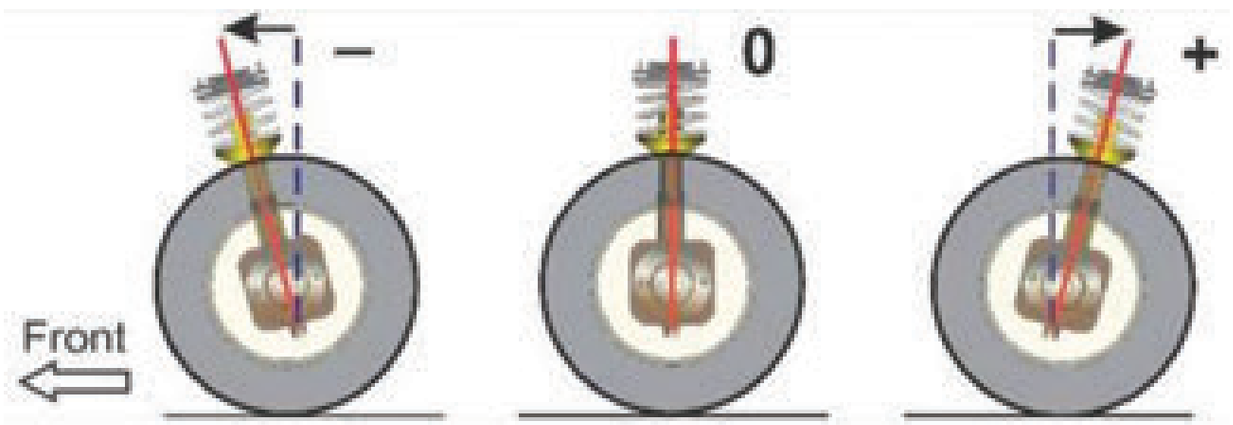


Figure 8.9.2.1 (c) Caster angle

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